

STATS 607 Project 1 Reflection

Originally, the analysis was split into three separate Jupyter notebooks. There are many duplicated code across the three notebooks, which makes the notebooks difficult to read and maintain. Also, there was not any file that introduces the project purpose or structure, nor were there docstrings that explain what the functions and classes in the notebook accomplish. Except these, the original state of the analysis also hinders the reproducibility of the analysis. In particular, the original analysis did not record the computing environment used for the analysis, and it did not fix random seeds prior to analysis.

One major challenge during the transformation was that fitting Jupyter notebook into a clean directory structure requires a complete reorganization of the code. Mere copying and pasting do not work. Therefore, I needed to write a few new functions based on the existing code and carefully manage the dependency among the functions to make sure that the final analysis runs properly. This also took some debugging efforts. Another challenge during the transformation was that, since the original code lacks proper documentation, I could not remember what some functions or analyses intend to accomplish. As a result, while reorganizing the code and writing documentation, I had to read the code in more detail to infer its functionality.

Two changes had the largest impact on reproducibility. First, I added code at the beginning of my analysis to fix the random seed. Second, I recorded the dependency versions of my analysis in the file “requirements.txt” and included the environment setup instructions in the README, so that other interested readers can recreate the computational environment.

For a future project, I will try to organize my code in a clear, modular directory structure to improve readability and maintainability. I will also write detailed documentation to make sure that myself and other readers can understand the code. Finally, I will fix random seeds for my analyses and record the environment setup to ensure reproducibility.

Roughly, reorganizing the code into the directory structure took me about four hours. Writing the test cases and the Makefile each took me about half an hour. Writing docstrings and the README took me about three hours.

I used AI to help me understand the project requirements, debug my code, and help with documenting the code. I reviewed AI’s outputs, and I took AI’s suggestions only when I understand them and feel confident that they are correct.